HighFinesse
The Standard of Accuracy

Linewidth Analyzer LWA-10k/100k

Manual

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1 Introduction

The HighFinesse Linewidth Analyzers of the LWA-10k/100k series are high-end optical instruments for measuring and analyzing the lineshape and frequency noise density spectra of narrow cw laser light sources.

A combination of an interferometric based measurement principle with specifically tailored ultra-low noise electronics enable the LWA-10k/100k to measure the laser frequency changes as a function of time.

Sophisticated algorithms incorporated in an easy-to-use software provide the user with the frequency noise density and lineshape of the laser under test.

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2 Unit description and Installation

2.1 Package Content

Scope of delivery for standard package:

- 1x LWA-10k/100k Analyzer Unit
- 1x LWA-D Digitizer Unit
- 1x BNC cable with T-adapter and $50\ \Omega$ termination plug
- 1x Ethernet cable
- 1x USB 3.0 cable (A male to B male)
- 1x Power cord (Analyzer Unit)
- 1x USB stick with software, manual and certificate

Please check for the completeness of the delivery before setting up the units.

2.2 Safety Instructions and Warnings

Before using the LWA-10k/100k please read and follow the following Safety Instructions and Warnings.

The following safety terms are used in this manual:

DANGER! ⚠	This symbol in this manual should draw attention to immediate danger that could result in personal injury or death.
CAUTION! ⚠	This symbol in this manual should draw attention to possible resulting hazards that could result in personal injury or death.
ATTENTION! ⚠	This symbol in this manual should draw attention to hazards that could damage the instrument.
NOTE! ⓘ	This symbol in this manual should draw attention to information that may be beneficial in the use of the instrument.

2.2.1 General Safety Terms

DANGER! ⚠	Before operating the LWA-10k/100k please read this manual carefully to avoid personal injury
CAUTION! ⚠	Prevent personal injury and damage to the electronics. The following safety instructions must always be followed.
CAUTION! ⚠	Wherever this symbol is attached, read and understand the manual before operating the instrument. The manual must be consulted in order to find out about the potential **HAZARDS** leading to personal injury or death and how to avoid them.
ATTENTION! ⚠	Wherever this symbol is attached, read and understand the manual before operating the instrument. The manual must be consulted in order to find out about the potential **HAZARDS** for the equipment and how to avoid them.
DANGER! ⚠	Do not position the instrument in a way that it is difficult to disconnect it.

DANGER! ⚠ If the instrument is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.

DANGER! ⚠ For safe operation and for proper grounding, please use the supplied mains cable only. Improper or missing grounding can lead to serious injury.

DANGER! ⚠ Operation of this electrical equipment in outdoor areas, wet or dirty atmosphere can cause unwanted behavior.

CAUTION! ⚠ Potential explosive atmosphere, corrosive atmosphere, and other environments that deviate from normal environmental conditions (IEC 61010-1 clause 1.4.1) are not allowed.

DANGER! ⚠ Do not open the LWA-10k/100k and Digitizer during operation. Only authorized and specially trained service personnel should maintain and repair components. Never open the internal power supply. To make sure that there is no mains connection, switch off the power switch on the rear side and remove the power cord. The button on the front panel is a switch for starting the device only.

CAUTION! ⚠ Do not cover or use the instrument outside the specified operation conditions.

2.2.2 Safety Features

2.2.2.1 AC Mains Voltage and Fuse

CAUTION! ⚠ Do not open up the instrument, the access to the fuse is on the back panel of the instrument. Disconnect the instrument from power before exchanging fuses if necessary.

ATTENTION! ⚠ The instrument is designed to operate on the mains voltage of either $115\ \mathrm{V}$ or $230\ \mathrm{V}$. Fuses have to be exchanged when the supply voltage is changed from $115\ \mathrm{V}$ or $230\ \mathrm{V}$ or vice versa.



Figure 1: The mains voltage setting is changeable by rotating the voltage selector in the IEC filter by 180°, to either the side with the $115\ \mathrm{V}$ marking or the $230\ \mathrm{V}$ marking.

2.2.2.2 Power Supply Specifications

Input Voltage	$115\ \mathrm{V}$ or $230\ \mathrm{V}$

Input frequency	$50\ \mathrm{Hz}$ / $60\ \mathrm{Hz}$
Maximum input power	$100\ \mathrm{W}$
Protection	Short circuit protection on all outputs, Over voltage protection, Overpower protection
Fuse for 230V	T250V - $3.15\ \mathrm{A}$
Fuse for 115V	T250V - $6.3\ \mathrm{A}$

2.2.2.3 Laser Safety Features

CAUTION! ⚠ The LWA-10k/100k is designed to handle laser power up to $20\ \mathrm{mW}$. No laser light or laser stray light can leave the LWA-10k/100k enclosure when used as intended (fiber connection correctly attached).

ATTENTION! ⚠ Do not exceed an input power of $20\ \mathrm{mW}$ to prevent damage to the analyzer.

2.3 Operating Principle

- Measurement of frequency deviation by frequency discrimination approach
- Direct evaluation of frequency noise density and optical lineshape spectrum
- Intrinsic (Lorentzian) and effective (optical) Linewidth evaluation

The spectral properties of laser light can be described in terms of the linewidth of the optical line shape spectrum or in terms of the frequency noise density distribution. Since the lineshape and the frequency noise evaluation approach in characterizing laser light are complementary, the access to each information allows one to determine valuable properties of the laser, which are necessary for multiple applications.

Both properties, the lineshape and the frequency noise spectrum, are related to the laser electric field:

$$E(t) = E_0 e^{i(2\pi\nu_0 t + \phi(t))}$$

where the phase φ(t) can be expressed as the temporal integral of the frequency deviations Δν(t) around its average value ν_0.

The frequency noise density spectrum gives information about the spectral distribution of the frequency noise on the laser and is thus the density spectrum of the frequency deviations measured in time, while the optical lineshape spectrum is the power spectrum of the laser light field. It can be seen, that access to both representations requires the measurement of the frequency deviations Δν(t).

The LWA-10k/100k is an instrument capable of measuring the frequency noise, whereas the measurement principle bases on a frequency discriminator that optically transfers frequency deviations of the laser light into intensity deviations. If the discriminator function from frequency deviation to intensity is well characterized, one can easily transfer the intensities (measured in time by a photodetector) back to frequency deviations.

This approach varies significantly from other techniques such as the self-heterodyne or self-homodyne techniques that give an optical spectrum of a beat-note depending mainly on the optical properties of the beating light fields. Generating the original spectrum of the pure light field from the beat note is thus non-trivial and only possible for certain cases. This can lead to erroneous values of the linewidth especially for a laser with a complex noise spectrum.

2.4 Linewidth at different observation times

Depending on the target application of the laser, the usage of two distinct properties related to the phrase “Linewidth” has been established, which we distinguish as “Intrinsic Linewidth” and “Effective Linewidth”. The “Intrinsic Linewidth” is a measure for the frequency noise at high frequencies of the laser (at an observation time of $1\ \mu\mathrm{s}$, which value is of special interest for high speed modulated laser manufacturers or for telecommunication techniques. The “Effective Linewidth” on the other hand is more interesting for spectroscopic applications e.g. in quantum optical applications. The observation time is given by the inverse of the resolution selected by the user and is stated next to the effective linewidth value in the software. See Section 5 for details.

2.5 Unit Description

2.5.1 Analyzer Unit

The Analyzer-Unit contains the optical elements of the LWA-10k/100k and is designed for a specified wavelength range. The Analyzer-Unit transforms frequency or intensity noise of laser light sources into analogue voltage noise, which can be further used for noise analysis or for noise reduction in combination with a feedback loop.

CAUTION! ⚠ Maximum input power $20\ \mathrm{mW}$!

NOTE! ⓘ The BNC jack of output 4 (MAIN) is not connected to the housing. Avoid short circuits to other potentials.



Figure 2: The Analyzer Unit

2.5.1.1 Front Panel

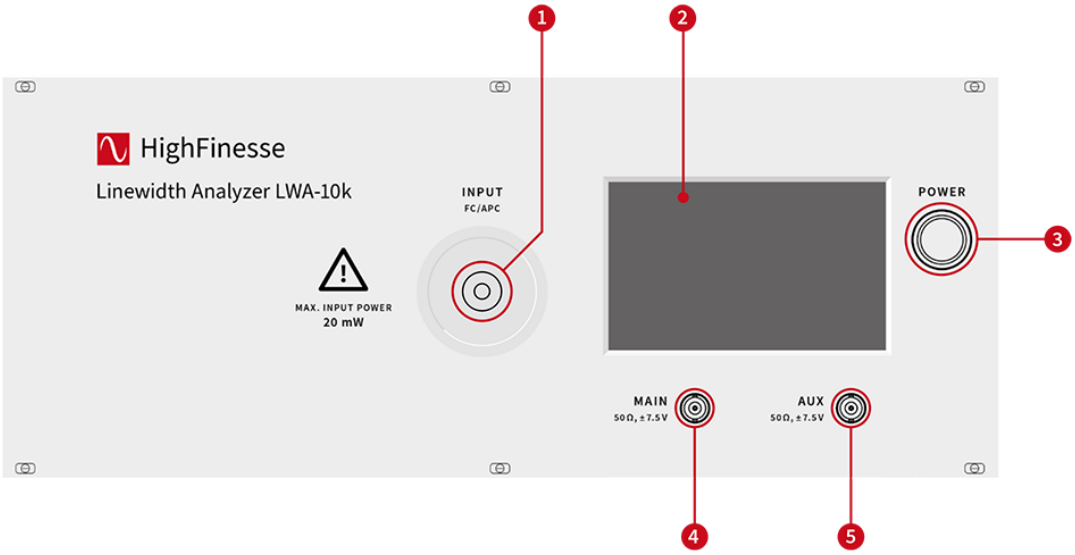


Figure 3: The front panel of the Analyzer-Unit, see Table 1 for descriptions.

Table 1: Functionality of the Analyzer Unit, see Figure 3 for positions.

Number	Function	Description
1	Input	Fiber-input with FC/APC connector.
2	Display	Indicating the current state of the instrument. See Section 2.5.1.2 for details.
3	Power Button	Button to set the device in operation state.
4	Main	BNC connector, analog output voltage: $\pm 7.5\ \mathrm{V}$ $(50\ \Omega)$. The analog output voltage is directly proportional to laser frequency or intensity noise.
5	Aux	No function. Res

DANGER! ⚠ The Power Button is not separating the instrument from the power supply. Please use the switch on the rear and disconnect the power cord to make sure the power is cut.

2.5.1.2 Display

The display indicates the current state of the instrument, the optical input power and the stabilization state. Please see Figure 4.

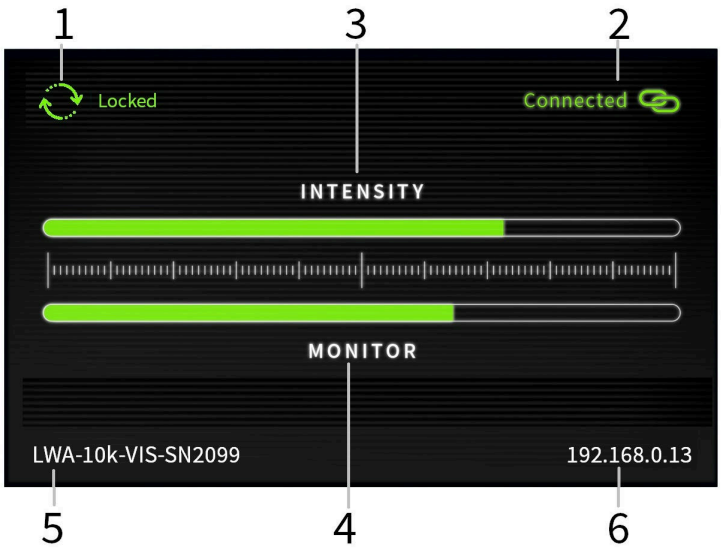


Figure 4: The display of Analyzer-Unit, see Section 2.5.1.2 for descriptions.

(1) Device status indicator

Indicates the current state of the Analyzer in its initialization procedure, one of “Warm Up”, “Adapting”, “Stabilizing”, “Locked”, and “Failed”. Wait for this indicator to turn to “Locked” before valid measurements can be obtained. If “Failed” is displayed or “Stabilizing” for more than 20 minutes refer to the troubleshooting Section 6.

(2) Connection status of the PC software

Indicates whether the instrument is recognized by the LWA control software (possible states: “connecting” and “connected”). Once “connected” is displayed the LWA is recognized by the software.

(3) Intensity bar

Serves as a measure of the relative power coupled to the instrument. Once green the power requirement are fulfilled to obtain valid measurements. If strong fluctuations are observed check the power stability of your laser and verify that an isolator is used between laser and LWA.

(4) Monitor

Serves as measure of the frequency noise of the laser. Once green the LWA has stabilized on the laser and the frequency stability criteria on the laser is fulfilled such that valid measurements can be obtained. If it does not turn green check the frequency stability of your laser. The frequency drift needs to be smaller than the specified maximum frequency stroke, See Section 7.2.

(5) IP address of the analyzer unit in the network

(6) Alarm icons

Indicate improper input power exceeding the maximum and minimum input power. See Section 7.2.

2.5.1.3 Rear Panel

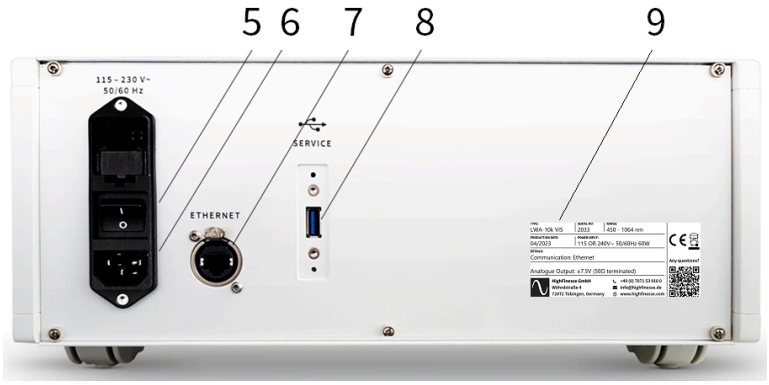


Figure 5: The rear panel of the Analyzer-Unit, see Table 2 for descriptions.

Table 2: Functionality of the Analyzer Unit, see Figure 5 for positions.

Number	Function	Description
5	Power Line Switch	Switch to enable or disable the power supply of the Analyzer-Unit.
6	Power Connection	C13 power connector for use with C14 power cable, as delivered with the instrument. Always use high-quality, standard-compliant power cords.
7	Ethernet interface	Ethernet RJ45 socket for use with RJ45 Ethernet cable, as delivered with the instrument. This supplies the communication interface between the Analyzer-Unit and the user's computer (Windows 10 or higher®) via direct connection or network.
8	Service port	The service port is not intended the use by the HighFinesse service Team and is closed with a metal cap.
9	Label	The label is shown in Figure 6.

ATTENTION! Do not remove the metal cap.

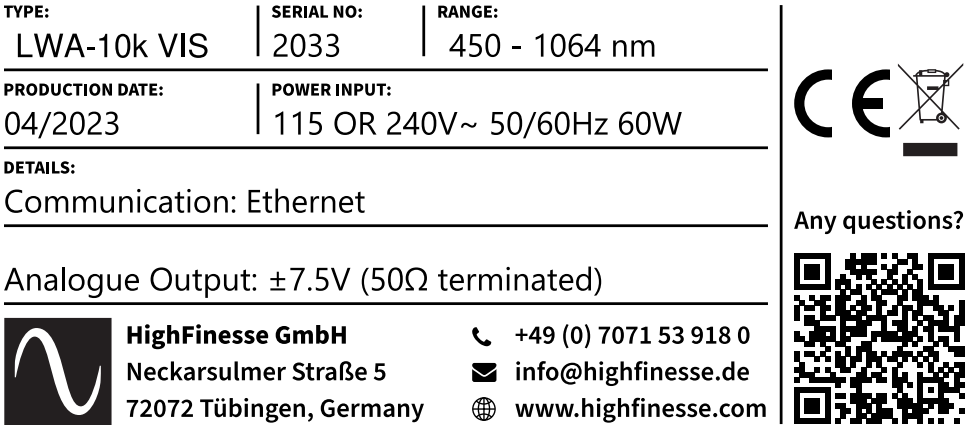


Figure 6: The label has the size 74 × 32 mm and colour silver/grey

2.5.2 Digitizer

The Digitizer digitizes the analogue output voltage of the Analyzer-Unit. The Digitizer is compatible with all Linewidth Analyzers from HighFinesse. It requires connection to both, the Analyzer-Unit and a computer with installed software. If the Digitizer is connected via an USB-Hub, ensure using one with external power supply.

2.5.2.1 Front Panel

Table 3: Functionality of the Digitizer Unit, see Figure 7 for positions.

Number	Function	Description
1	Main	BNC connector, analog input voltage maximum ±20 V (range is set by GUI). Connect to the “Main” Output of Analyzer-Unit via T-adapter with 50 Ω termination. Place the T-adapter with 50 Ω termination directly at the Main input of the Digitizer.
2	Aux	No function. Reserved for future use.
3	Trigger	No function. Reserved for future use.

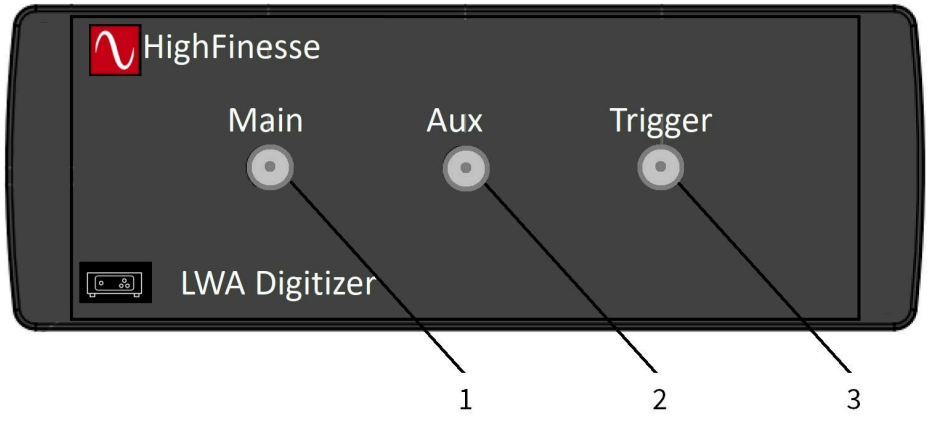


Figure 7: The front panel of the Digitizer-Unit. Input BNC connector: analog input voltage maximum ±20 V.

2.5.2.2 Rear Panel

Table 4: Functionality of the Digitizer Unit, see Figure 8 for positions.

Number	Function	Description
1	Label	The label is shown in Figure 9.
2	USB Type B socket	USB 3.0 Type B Connection for transfer of digitized data to the computer.



Figure 8: The rear panel of the Digitizer-Unit. USB 3.0 Type B Connection for transfer of digitized data to the computer.

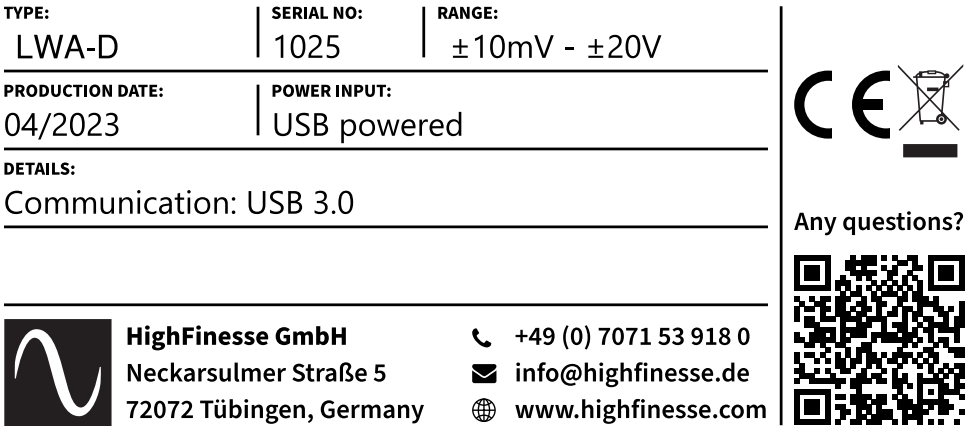


Figure 9: The label has the size 74 × 32 mm and colour silver/grey

3 Getting Started

Safety regulations:

The usage of the instrument is restricted to authorized personnel, who have familiarized themselves with the content of this manual and the operation of the instrument. Methods of operation that compromise safety must be avoided.

The user must inspect the device for visible defects before operation to ensure it is in perfect condition before using it.

Do not remove or disable any security features except when needed for repairs or maintenance. Make sure that all security features are reinstalled as soon as the work is completed.

Unplug the instruments power cord before maintenance as described in Section 3.2.3.

It is strictly forbidden to modify the instrument.

3.1 Requirements on the laser for a successful measurement

- The laser must be cw and singlemode.
- The laser must not be scanned during the measurement.The maximum frequency stroke specified must not be exceeded. See Section 7.2.
- The laser must not exceed the maximum linewidth specified. See Section 7.2.
- In case the laser is sensitive to back reflections, the use of an isolator with $55\ \mathrm{dB}$ between laser and LWA is recommended.

3.2 Hardware Installation

- Inspect the package content after delivery for any damage.
- Check if the setting at the Analyzer-Unit rear panel matches the correct voltage for your power grid.
- Connect the Analyzer-Unit via Ethernet cable (rear) to the network or directly to the computer and via C14 power cord to the power grid. Please see Figure 10.
- Connect the laser light source to the Analyzer-Unit via a polarization maintaining (PM) fiber with FC/APC connectors to the system.

CAUTION! ⚠ The LWA-10k/100k is designed to handle laser power up to $20\ \mathrm{mW}$. No laser light or laser stray light can leave the LWA-10k/100k enclosure when used as intended (fiber connection correctly attached).

ATTENTION! ⚠ Do not exceed an input power of $20\ \mathrm{mW}$ to prevent damage to the analyzer.

- The correct optical power level is a necessary condition for a successful initialization procedure. Please adjust the power to the recommended values specified in Section 7.2.
- Switch on power on the back side of the Analyzer-unit.
- Switch on instrument on the front side of the Analyzer-unit.
- See Section 2.5.1.1 for interpreting the Analyzer-Unit front panel display.

- At start-up, the instrument may need up to 3 minutes for initialization. During this period, the system is tuned to working conditions.
- Setup the corresponding Digitizer. Connect the Digitizer via USB to a computer. Connect the Digitizer to the Analyzer “Main” output via BNC-cable with $50\,\Omega$ termination at the Digitizer “Main” input.

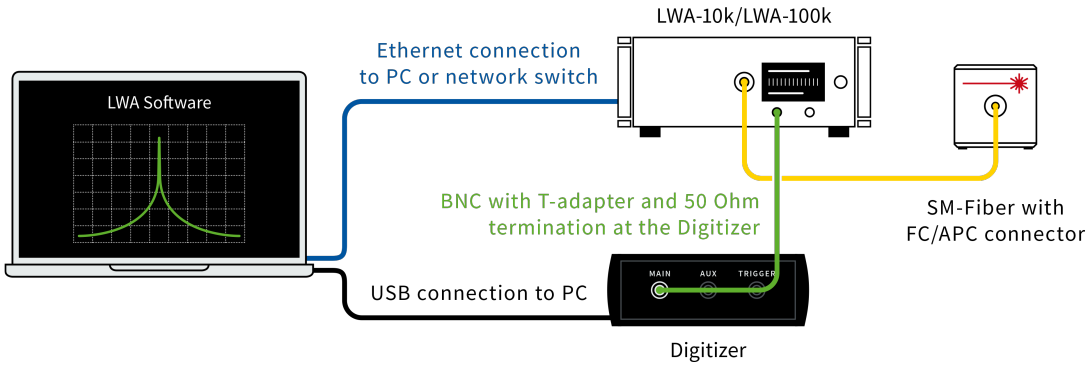


Figure 10: Interfacing the Analyzer and Digitizer

3.2.1 Main Power Switch and Initialization

When switched on via the power switch (rear side of Analyzer) and the power button (front side), the system performs an initialization procedure, during which the optical system is adjusted to the laser under test.

If the optical input level is in the required range the power level indicator shown on the display turn green.

Depending on ambient conditions, the initialization process may take up to 3 minutes. This initialization status is indicated the icon on the upper left of the display. Successful completion of initialization and stable operation of the system is indicated by the status icon and the green color of the monitor bar on the display. In this status the system output (4) is unmuted automatically.

In the unmuted state, the analogue output is now active and providing a voltage proportional to the laser frequency/intensity noise.

3.2.2 Recommendations for a successful Initialization Procedure

For a successful initialization procedure it is highly recommended not to alter the laser frequency, the optical output power or the laser polarization, during the initialization. The optical fiber should be fixed on an optical table that is not affected by vibrations. Ideally a polarization maintaining fiber is used.

The state of the initialization procedure is indicated by the icon on the upper left on the display. The optical power indicator bar should be green. The instrument is preparing itself to stabilize on the laser light. During the procedure the monitor bar below the power indicator bar is changing until the desired state is reached.

If initialization endures longer than 5 minutes, please contact HighFinesse service. The initialization is finished if the icon shows “Locked” and the monitor bar turns green.

Excessive changes in laser frequency by more than the specified maximum frequency stroke, optical power or laser light polarization may cause the instrument to reset to initialization procedure. Small changes in frequency, power or polarization can be compensated whereas the current measurements might be invalid (see Section 4).

3.2.3 Maintenance

Routine inspection is not required when the instrument is used under normal working conditions. Do not attempt to open the housing. There are no user-serviceable parts inside and the warranty is void, if the seal is broken. If the device appears to be working incorrectly, check the following:

- Inspect the device for scratches, dents or signs of damage due to handling.
- Verify that the housing has not been opened.
- Check the power source connection.
- Ensure that the operating environmental conditions are within specifications.
- Is the laser equipped with an optical insulator? Is the laser running on a single longitudinal mode? Is there excessive RIN on the laser? Can you exclude frequency changes higher than the specified maximum frequency stroke (See Section 7.2)?

If the solutions suggested above do not solve your problem, please contact HighFinesse service for further assistance.

3.3 Software Installation and Start

Recommended specifications for a suitable computer are:

- Windows 10, 64 bit or higher
- Intel® Core™ i5 equivalent or better
- Memory 16 GB or better
- At least one USB 3.0 and one Ethernet port

Insert the USB flash drive attached to this manual into the target computer.

Run the installation executable in the USB root directory as administrator (necessary for installing the required drivers). It is recommended to install the software to a folder with write permission.

Before starting the software ensure that both the Analyzer and Digitizer are connected to the computer or network, the Analyzer is powered up and the Digitizer is connected to the Analyzer using the BNC cable.

Start the software by running the “LWA Control” executable.

Before starting any measurement, it is strongly recommended to await the end of the initialization procedure of the Analyzer Unit (Analyzer front panel display and GUI status panel, monitor and power).

3.4 Minimization of External Noise Sources

Both, the Analyzer and the Digitizer are highly sensitive instruments designed to operate with extremely low voltage noise. In order to avoid external noise sources affecting the measurement by unwanted ground loops we recommend the following setup:

1. Computer or Laptop directly connected to Analyzer via Ethernet cable with non-metallic housing (no shielding, UTP type) to avoid ground loops
2. Direct connection between Digitizer and Computer via USB 3.0 cable
3. No further devices connected to Computer to avoid additional noise

4 LWA Control Software

Before starting the software make sure the IP address is shown on the display of the analyzer unit to verify that the Ethernet connection is established. In addition, check that the digitizer is connected via USB.

4.1 Instrument Selection

Once started via the shortcut created on the Desktop the software will search for all connected instruments. Once the search finished, the instrument selection screen, see Figure 11, will appear. To continue, select the desired instruments and press the “Connect” button. If the desired instrument is not shown, make sure the Unit is connected to the computer and is turned on. Hit “Restart Search” to rescan for new devices.

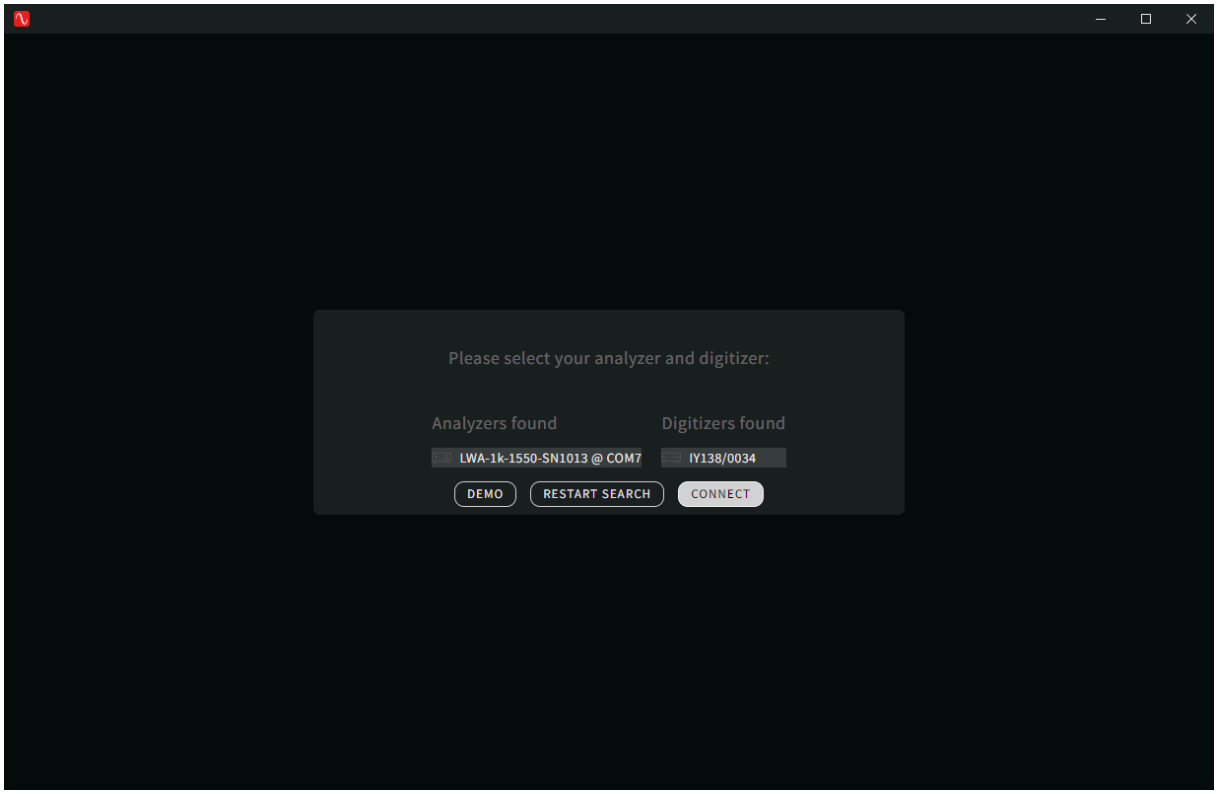


Figure 11: The instrument selection screen of the software.

4.2 Main Interface

Once Analyzer and Digitizer are selected, the software will transition to the main interface, as shown in Figure 12. On the left side the instrument status and the measurement settings are located, for more information see Table 5. The evaluated data in shown in different tabs, see Section 4.2.3 for more info. These tabs can be freely rearranged using drag and drop.



Figure 12: The main interface if the software, see Table 5 for descriptions.

Table 5: Description of the software main interface, see Figure 12

Number	Function	Description
1	Digitizer Status	Shows the connection status of the digitizer unit.
2	Analyzer Status	Shows the connection and initialization status of the analyzer unit. Restart the analyzers initialization procedure if needed by clicking on “Reset”.
3	Measurement Settings	see Section 4.2.2
4	Optical Input Control	This displays the estimated optical input power and relative transmission through the interferometer.
5	Data Charts	The captured and evaluated data is displayed in interactive charts. Available charts are: Time Series, Frequency noise density, Lineshape and Trends showing the longterm development of the evaluation results (6)
6	Result panels	Displays evaluation results based on the dataset shown in the corresponding chart. The analysis leading to the results is described in Section 5.

4.2.1 Main Menu

The main menu is located in the title bar.

- LWA Control:

  - Manage Instruments: This lets you go back to the instrument selection dialog, for switching to another instrument. See Section 4.1.
  - Quit: closes the software
- File: file related operations
  - Export: opens the export dialog, see Section 4.2.5.
- Display: customize the visual appearance of the software.
  - Light/Dark theme: switches from a darker to a lighter color scheme, respectively.
  - Small/Large font: Adjusts the font size of all elements.
  - Small/Large results: Adjusts the font size of the result panels.
  - Compact/Detailed: Loads the corresponding layout preset. All tabs can be individually repositioned by dragging them around.
- Support: support related functionality
  - Contact HighFinesse: Automatically opens an eMail template to our service team.
  - About: Shows software version information, helpful to include in service cases.

4.2.2 Instrument Information Panel

- Analyzer/Digitizer connection: shows whether the unit is still connected.
- Analyzer status: Indicates the current state of the Analyzer in its initialization procedure, one of “Warm Up”, “Adapting”, “Stabilizing”, “Locked”, and “Failed”. Wait for this indicator to turn to “Locked” before valid measurements can be obtained. If “Failed” is displayed refer to the troubleshooting Section 6.
- Data acquisition running/paused: indicates the current data acquisition status
- Wavelength: It is needed to specify the current input wavelength with an accuracy of $10\ \mathrm{nm}$ here to obtain accurate results.
- Resolution: Sets the perturbation frequency resolution of the frequency noise density. This corresponds to changing the observation time to the inverse of the perturbation frequency resolution.
- Average Mode: Changes the averaging behaviour.
  - Off: No averaging performed.
  - Fixed *N*: Averages the last *N* samples. All *N* samples can be exported. *N* can be 10 at maximum.
  - Continuous: Averages continuously until stopped. Only the final result can be exported, individual time series are not available.
- Optical Power: Displays the estimated optical input power.
- Monitor: Shows the relative transmission through the interferometer.

4.2.3 Chart Tabs

Shows the respective dataset. The value of the mouse cursor position is shown on the lower left. If the initialization procedure has not finished or the lock is interrupted due to large laser frequency/power changes the shown dataset is colored gray, signaling a less-trustworthy measurement. Interactions with the charts can be customized using the toolbar in the top right, see Figure 13.

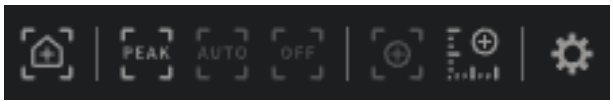


Figure 13: The chart toolbar

From left to right the groups of the toolbar have the following functionality:

- Home: reset the plot to the default view
- Autoscaling behavior:
  - peak: The y axis is automatically resized to the largest and smallest observed values.
  - auto: The y axis is automatically scaled to fit the current dataset.
  - off: The axis is not adjusted automatically.
- Zoom modes:
  - box zoom: This zooms into the drawn box region, disabling any autoscaling.
  - directional zoom: This zooms in either x or y direction only, preserving autoscaling if possible.
- Settings: Chart specific settings

Below each chart, you will find evaluations based on that dataset. These are based on the last dataset. If averaging is active the averaged dataset is analyzed. An exception is the time series dataset for which always the latest set is analyzed.

4.2.4 Chart Settings

Click on the gear icon in the top right corner of the chart to modify its settings. Some options are marked with ⃠ on hover and cannot be changed. These are reserved for future software updates.

4.2.4.1 “Frequency Noise Density” Chart

Figure 14 shows the settings for the Frequency Noise density chart and results panel.

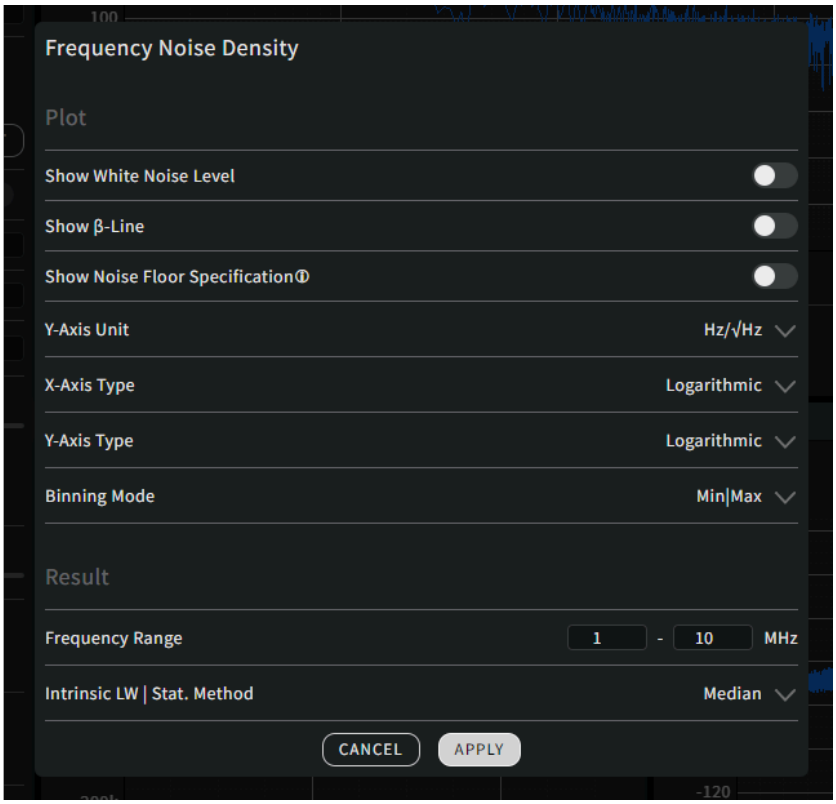


Figure 14: The frequency noise density settings dialog

- Show White Noise Floor: Shows or hides an indicator line for the white noise floor of the FND ($> 1\,\mathrm{MHz}$)
- Show β-line: Enables or disables the β-separation line
- Show Noise Floor Specification: Shows or hides an indicator line for the instruments noise floor specification.
- Y-Axis Unit: Decide if FND or PND is displayed. Choose between $\mathrm{Hz}/\sqrt{\mathrm{Hz}}$ (FND) and $\mathrm{rad}/\sqrt{\mathrm{Hz}}$ (PND). See also section Section 5.2 and Section 5.3.
- X/Y-Axis Type: Allows switching between logarithmic and linear axis
- Binning Mode: Changes the binning mode for displaying the large dataset. As a result of the large dataset and the finite screen resolution, not all points can be displayed which is why a representative point per bin size is displayed. Currently the binning is fixed to displaying the maximum and minimum point per bin.
- Frequency Range: Adjusts the range taken in account for intrinsic LW evaluation (currently not available)
- Intrinsic LW | Stat. Method: Adjust the statistical method used to evaluate (currently not available)

4.2.4.2 “Time Series” Chart

Figure 15 shows the settings for the time series chart and results panel.

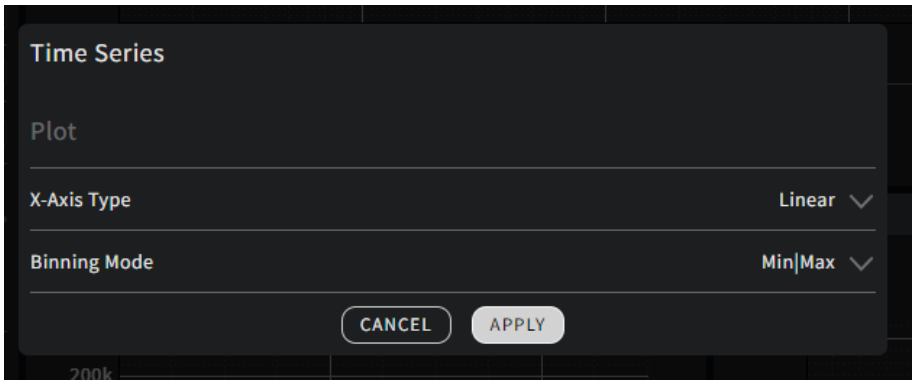


Figure 15: The time series settings dialog

- X-Axis Type: Allows switching between logarithmic and linear axis
- Binning Mode: Changes the binning mode for displaying the large dataset. As a result of the large dataset and the finite screen resolution, not all points can be displayed which is why a representative point per bin size is displayed. Currently the binning is fixed to displaying the maximum and minimum point per bin.

4.2.4.3 “Lineshape” Chart

Figure 16 shows the configurations options of the lineshape plot and results panel.

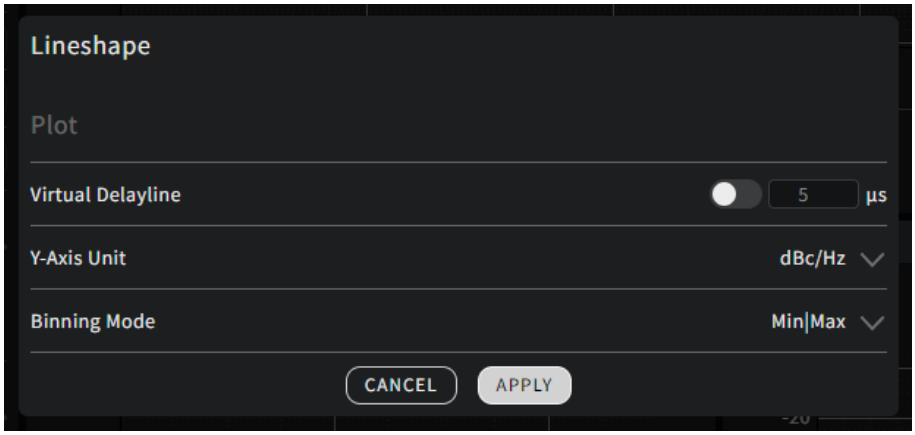


Figure 16: The lineshape settings dialog

- Virtual Delayline: Enables simulation of potentially distorted lineshape as it would be measured by a delayline setup (see Section 5.4.2). This should be deactivated and used for comparison to these setups only.
- Y-Axis Unit: Changes the lineshape unit. Currently fixed to $\mathrm{dBc/Hz}$
- Binning Mode: Changes the binning mode for displaying the large dataset. As a result of the large dataset and the finite screen resolution, not all points can be displayed which is why a representative point per bin size is displayed. Currently the binning is fixed to displaying the maximum and minimum point per bin.

4.2.5 Export Dialog

The export dialog can be accessed though the main menu “File → Export” or the “Ctrl + S” shortcut. It is shown in Figure 17. It allows customization of the exported data.

The availability of the datasets does depend on the currently active averaging mode:

- “Average Mode” set to “Off”: all data is available, the current sample is exported
- “Average Mode” set to “Fixed”: everything is available, time series data sets are exported individually, The resulting averaged FND and Lineshape are exported.
- “Average Mode” set to “Incremental”: only the averaged FND and Lineshape data sets are available, time series data sets are not available and will we ignored

It is possible to save as `numpy`s compressed binary format `npz` and the text based `csv` format.

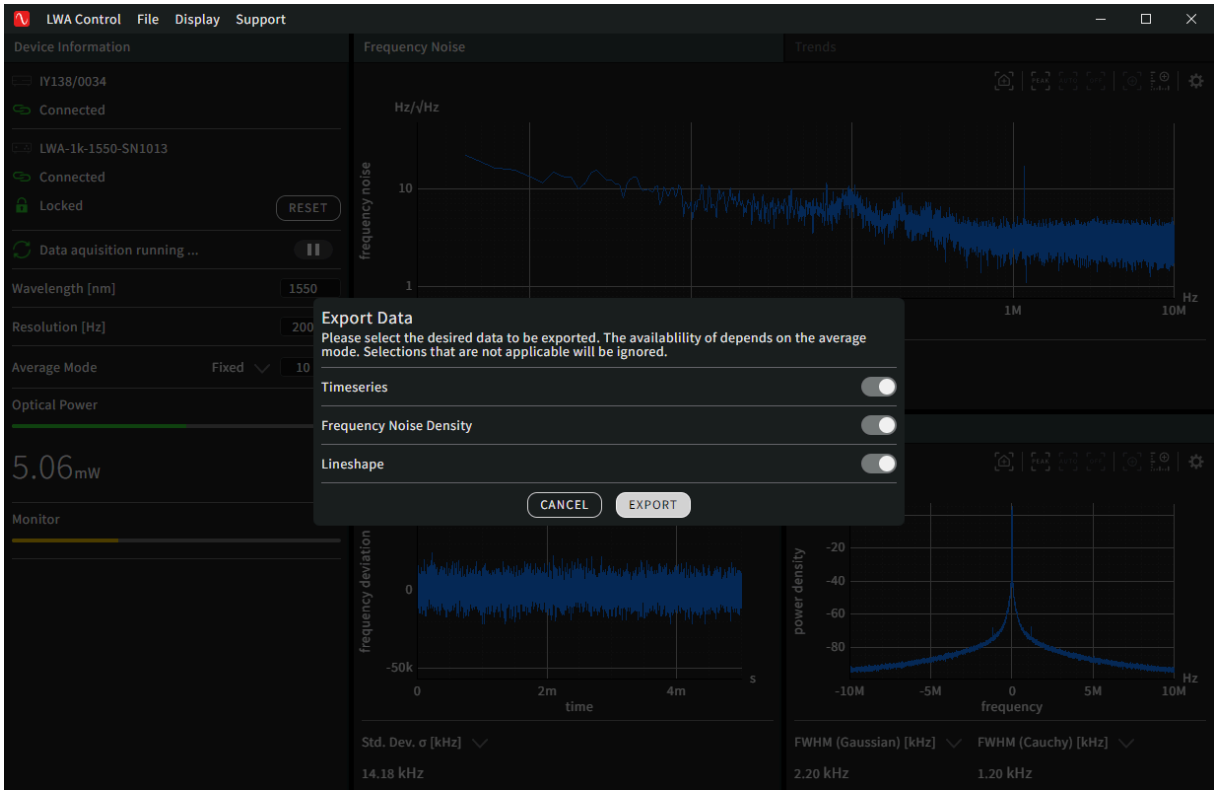


Figure 17: The data export dialog

5 Linewidth / Frequency Noise Analysis

5.1 Frequency Deviations in Time

NOTE! ⓘ This is the fundamental dataset from which all further quantities are derived from.

The operating principle of the LWA is the recording of analogue voltage signals as a measure for the frequency deviations in time domain. By the knowledge of the carefully characterized frequency discriminator and the optical power, the analogue voltage is transferred into frequency deviations Δv(t) [Hz] per digitizer timebase.

From this fundamental dataset many quantities of interest can be derived without approximation (some of them are automatically calculated by the LWA Control Software). The LWA Control Software automatically calculates the frequency noise density spectrum S_N $[\mathrm{Hz}/\sqrt{\mathrm{Hz}}]$ by performing a Fourier Analysis, of Δv(t)

$$S_N(f_P) = \sqrt{\mathcal{F}(\Delta\nu(t)) \times \mathcal{F}^*(\Delta\nu(t))}.$$

The conversion to phase noise density P_N $[\mathrm{rad}/\sqrt{\mathrm{Hz}}]$ is straight forward:

$$P_N(f_P) = \frac{1}{f_P} \times S_N(f_P).$$

Here f_P is the perturbation frequency (Fourier frequency).

In addition, the lineshape spectrum S_L is obtained by a Fourier analysis of the electric fields derived from Δv(t):

$$S_L(f) = \mathcal{F}\left(e^{2\pi i \int \Delta\nu(t)\,\mathrm{d}t}\right) \times \mathcal{F}^*\left(e^{2\pi i \int \Delta\nu(t)\,\mathrm{d}t}\right)$$

5.2 Frequency Noise Density Spectrum (FND)

The LWA Control Software automatically calculates the frequency noise density spectrum S_N $[\mathrm{Hz}\ /\sqrt{\mathrm{Hz}}]$ by performing a Fourier Analysis, of Δv(t)

$$S_N(f_P) = \sqrt{\mathcal{F}(\Delta\nu(t)) \times \mathcal{F}^*(\Delta\nu(t))}.$$

The Python example `timeseries-to-fnd.py` shows how to compute $S_N(f_P)$ based on a `npz` archive containing the timeseries (frequency deviations in time). See Section 4.2.5 for how to export data as `npz`.

- Spectral distribution of the laser frequency noise as a function of the perturbation frequency f.
- Enables assignment to noise sources (perturbations from power grid at the power grid frequency and higher harmonics, acoustic perturbations in the kHz range, etc.).
- Estimation of intrinsic (Lorentzian) linewidth at short observation times ($1\ \mu\mathrm{s}$) and effective (optical) linewidth at longer observation times (depending on resolution setting 1-$100\ \mathrm{ms}$) based on the β-separation line line method.

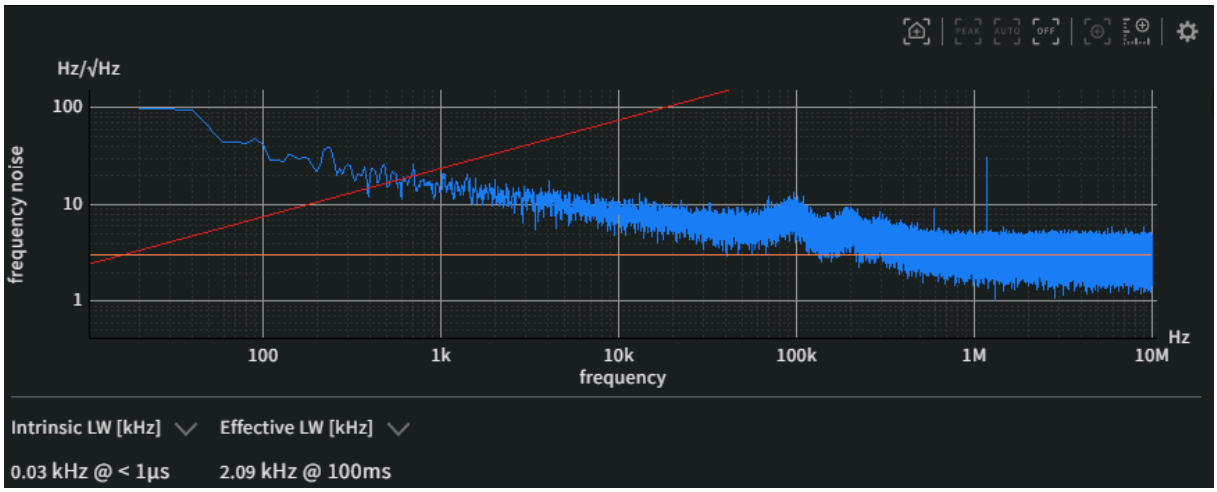


Figure 18: The frequency noise density spectrum with β-separation line (red) and white noise level (orange)

The frequency noise density spectrum may show the following features:

5.2.1 $1/f^{\alpha}$ – Decay

With increasing perturbation frequencies the laser frequency noise density decreases with $1/f$-behavior. Sources for $1/f$ noise can be various. Mainly it's related to so called burst noise or random telegraph noise which causes random jumps in frequency. Main reason for burst noise is based on trapping and releasing of charge carriers at thin films of electronic or opto-electronic devices. Additionally, higher order decays can occur, depending on laser design.

In addition, slow drifts or scans can contribute to the frequency noise in this low frequency regime.

5.2.2 Noise Floor

For high perturbation frequencies the laser frequency noise density may reach a constant value S_0. Above a certain perturbation frequency the laser frequency noise density is constant and thus represents the white noise component. Pure white noise results in a Lorentzian line in the optical spectrum whereas the full width at half maximum of the Lorentzian ν_{Lor} is given by:

$$\nu_{\mathrm{Lor}} = S_0^2 \pi$$

The noise floor of the frequency noise density spectrum can be determined either by a statistical analysis of the frequency noise density distribution at high frequencies or by a curve fitting procedure of the frequency noise density spectrum.

5.2.3 Additional Features

In addition to the $1/f$ decay and the noise floor at high frequencies the frequency noise density spectrum might exhibit peaks at characteristics frequencies allowing for a direct assignment to specific noise sources. An example for that is peaks at $50\ \mathrm{Hz}$/ $60\ \mathrm{Hz}$ and the

higher harmonics due to perturbations by the power grid acting on the power supply of the laser.

5.2.4 Estimation on Optical Spectrum

The shape and width of the optical laser spectrum depends on all frequency noise sources depending on their noise density and Fourier-frequency. For estimating the optical/effective linewidth sym.nu$_{opt}$ based on the frequency noise density, we use the so called β-separation approach[1)]:

$$\nu_{\mathrm{opt}} \approx \sqrt{8\ln(2)\int H\left([S_N(f)]^2 - \frac{8\ln(2)f}{\pi^2}\right) S_N^2(f)\,\mathrm{d}f}$$

with H being the Heaviside function and f the Fourier-frequency.

Graphically, all noise densities above the β-separation line are integrated. Therefore, the optical linewidth consists of the $1/f$-contribution (various statistics), other spikes and bands at lower frequencies and the noise floor offset (Lorentzian linewidth). Due to simplifications and assumptions, the β-separation method is an estimation for the effective linewidth. Therefore, evaluation of the optical spectrum is preferable for determining the effective linewidth, if the lineshape is accessible. This is the case via the direct calculation of the lineshape provided by the LWA Control Software as described in Section 5.4.

In general, on short timescales the laser shows a Lorentzian shaped spectral line, which linewidth is directly related to the white noise level. All frequency noise components with lower frequency affect this Lorentzian line which undergoes frequency changes in time.

5.2.5 Effective Linewidth (β-separation line)

- Combination of intrinsic linewidth and additional broadening mechanisms (thermal, electronical and acoustic noise).
- Determination by β-separation method (frequency noise density spectrum).
- Dominated by low frequency components and the $1/f$-part of the noise spectrum.
- Estimation for linewidth of optical spectrum at a specific observation time

5.2.6 Intrinsic Linewidth

- Limited by fundamental quantum processes and laser design.
- Determined by the noise floor level (white noise) of the frequency noise density spectrum and calculated by (rule of thumb): frequency noise density (in Hz^2/Hz) times π.
- Frequency noise level at high frequencies
- Lorentzian contribution to optical linewidth

[1)] Di Domenico et al., “Simple approach to the relation between laser frequency noise and laser line shape”, *Appl. Opt.*, vol. 49, no. 25, pp. 4801-4807, 1 Sep 2010.

5.3 Phase Noise Density Spectrum (PND)

The phase noise density spectrum (PND) P_N $[\mathrm{rad}/\sqrt{\mathrm{Hz}}]$ is calculated based on the FND S_N via the equation:

$$P_N(f_P) = \frac{1}{f_P} \times S_N(f_P).$$

Here f_P is the perturbation frequency.

5.3.1 Deriving PND from FND

Starting with the definition of the Fourier transformation (1) of the two quantities optical phase $\varphi(t)$ and optical frequency $\nu(t)$.

$$\begin{aligned} \varphi(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\varphi}(f) e^{2\pi i f t} \, \mathrm{d}f \\ \nu(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\nu}(f) e^{2\pi i f t} \, \mathrm{d}f \end{aligned} \tag{1}$$

The relation of these quantities is given by (2).

$$\nu(t) = \frac{1}{2\pi} \frac{\mathrm{d}\varphi(t)}{\mathrm{d}t} \tag{2}$$

Substituting (1) on both sides

$$\frac{1}{2\pi} \frac{\mathrm{d}\varphi(t)}{\mathrm{d}t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} i f \hat{\varphi}(f) e^{2\pi i f t} \, \mathrm{d}f = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\nu}(f) e^{2\pi i f t} \, \mathrm{d}f = \nu(t) \tag{3}$$

Comparing the two fourier integrals in (3) gives the following relation.

$$\hat{\nu}(f) = i f \hat{\varphi}(f) \tag{4}$$

The power spectral density is defined as (5).

$$\mathrm{PSD}_\nu(f) = \lim_{T \to \infty} \frac{1}{T} |\hat{\nu}_T(f)|^2 \tag{5}$$

Substituting (4) gives the relation of the PSDs.

$$\mathrm{PSD}_\nu(f) = \lim_{T \to \infty} \frac{1}{T} |i f \hat{\varphi}_T(f)|^2 = f^2 \lim_{T \to \infty} \frac{1}{T} |\hat{\varphi}_T(f)|^2 = f^2 \mathrm{PSD}_\varphi(f) \tag{6}$$

With $P_N(f) = \sqrt{\mathrm{PSD}_\varphi(f)}$ and $S_N(f) = \sqrt{\mathrm{PSD}_\nu(f)}$ this leads to the desired result:

$$P_N(f) = \frac{1}{f} \times S_N(f) \tag{7}$$

5.4 Lineshape or relative optical spectrum

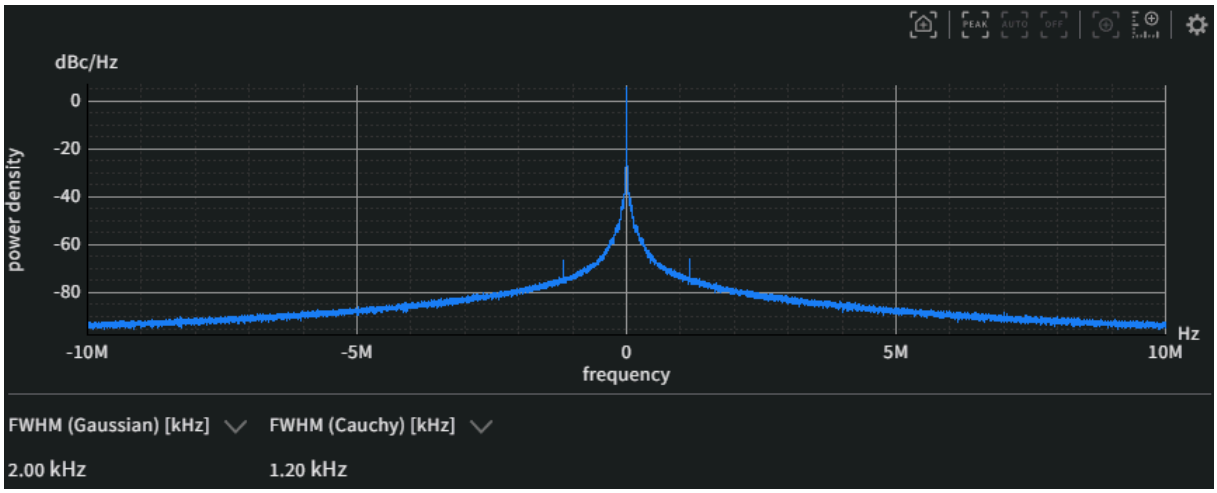


Figure 19: Example of a laser lineshape.

The lineshape or relative optical spectrum yields additional information that are not directly accessible in the frequency noise density spectrum:

- exact optical lineshape and linewidth
- possible sidebands

Based on the measured laser frequency deviations Δv(t) the lineshape/relative optical spectrum $\mathcal{L}(f)$ can be calculated via a Fourier Analysis of the electric fields $E(t)$ following the equations below:

$$\frac{E(t)}{E_0} = \exp\left(2\pi i \int_0^t \nu(u)\,\mathrm{d}u\right)$$

$$\mathcal{L}(f) = \mathcal{F}(E(t)) \times \mathcal{F}^*(E(t)).$$

The lineshape represents the real optical spectrum relative to the carrier frequency in power units. The Python example `timeseries-to-lineshape.py` shows how to compute $\mathcal{L}(f)$ based on a `npz` archive containing the timeseries (frequency deviations in time). See Section 4.2.5 for how to export data as `npz`.

It is possible to observe and analyze 1) sidebands, their separation to the carrier frequency and their suppression as well as 2) the lineshape, its symmetry, as well as underlying contributions and their contribution to the overall width.

A standard model for the laser lineshape is described by a Voigt-function containing the white noise component (Cauchy/Lorentzian) and the $1/f$- contribution (Gaussian). The LWA control software provides you with an “on the fly” estimation for a pure Cauchy/Lorentzian and Gaussian lineshape. In case you would like to fit a Voigt Profile, it is needed to export the data.

However, a real laser lineshape might deviate from this as it might be composed of multiple noise statistics or modulations that cannot be described by a simple model function any more.

5.4.1 Effective Linewidth (FWHM)

- Represents the linewidth of the optical spectrum
- The software calculates an estimation of the FWHM for the two cases purely Lorentzian and purely Gaussian. This gives a lower and upper bound on the linewidth. If the relative strength of both signals is desired, the data needs to be exported and analyzed externally. This can be done using the Python example described in Section 5.5.2.
- The FWHM of the lineshape is comparable to the effective linewidth determined by the β-separation method.
- The shape of the spectrum is determined by different statistical distributions causing the frequency noise with respect to the noise density.

5.4.2 Virtual Delayline Spectrum

For comparing the results of the LWA to the possibly distorted optical spectrum that can be obtained by a self-heterodyne delayline setup, the LWA control offers the opportunity to simulate the expected spectrum of a self-heterodyne delayline setup.

The procedure is based on evaluating the superposition of the calculated electric field with the same electric field shifted by the user given delay time τ.

$$S_{\mathrm{VDL}} = \mathcal{F}\left(e^{2\pi i \int \Delta\nu(t)\,\mathrm{d}t} + e^{2\pi i \int \Delta\nu(t+\tau)\,\mathrm{d}t+\phi}\right) \times \mathcal{F}*\left(e^{2\pi i \int \Delta\nu(t)\,\mathrm{d}t} + e^{2\pi i \int \Delta\nu(t+\tau)\,\mathrm{d}t+\phi}\right)$$

NOTE! ⓘ The resulting spectrum can show significant differences to the Lineshape spectrum. The simulated spectra might exhibit artificial distortions due to the drawbacks of the self-heterodyne delayline technology. This function should be used for direct comparison to these setups only. Whenever exact optical spectra are desired it should be disabled.

5.5 Python Examples

The Python examples can be found in the installation path under the `\examples\python` subdirectory. They use `npz` archives as input. See Section 4.2.5 for how to export data as `npz`.

5.5.1 Conversion Tools

The Python example `timeseries-to-fnd.py` shows how to compute $S_N(f_P)$ based on a `npz` archive containing the timeseries (frequency deviations in time).The Python example `timeseries-to-lineshape.py` shows how to compute $\mathcal{L}(f)$ based on a `npz` archive containing the timeseries (frequency deviations in time).

5.5.2 Lineshape Fitting Tool

The Python example `lineshape-fit.py` can be used to further analyze lineshape data exported as `npz` archive. It is demonstrated how to fit a Voigt model to the data and obtain Lorentzian and Gaussian linewidth as fit parameters.

6 Troubleshooting

This section list common questions for issues with LWA operation. In case a different error occurs or the provided steps do not improve the situation, please contact HighFinesse Service, see Section 9.

- Analyzer-Unit cannot finish initialization GUI does never show “Locked” or shows “Failed”
  1. Check if monitor bar is changing
  2. Check if laser if affected by strong noise sources, frequency drifts or non-single mode operation
  3. Check if optical power is stable
  4. Ensure stable ambient conditions
  5. Restart the Analyzer-Unit
- Analyzer-Unit permanently restarts the initialization procedure
  1. Laser frequency, power or polarization is not stable enough.
- Analyzer-Unit is not found by the software

1. Check if an IP address is displayed in the lower right corner of the Analyzer display
2. If this is not the case make sure that the Analyzer is connected to the same PC or network using the Ethernet cable and is turned on

- Digitizer-Unit is not found by the software
  1. Check if the devices are properly connected
  2. Reinstall the software to make sure drivers are installed
  3. Restart PC for proper driver initialization
  4. If a USB-Hub is used in combination with the Digitizer, try to directly connect the devices to the PC without the USB-Hub
  5. Check if your system matches the requirements
- Timeseries shows strange behavior
  1. Reconnect Digitizer-Unit
  2. Check BNC-T adapter and $50\ \Omega$ termination
- Linewidth is ultra-narrow
  1. Check if the Analyzer is connected and in operation mode (GUI and Analyzer-Front)
  2. Check that the Analyzer is connected to the Digitizer using the BNC cable
  3. Check for $50\ \Omega$ termination of the Digitizer.
  4. Check if the initialization procedure has been completed
- Measurements are evaluated really slow
  1. Check the “Resolution $[\mathrm{Hz}]$” setting. For lower resolutions the measurement time is increased as well as the size of the datasets. Increase the resolution for faster evaluation.
  2. Check if other resource consuming programs are running parallel to the control software
- Lineshape is not a symmetric peak

1. Check noise spectrum for large noise sources in the low frequency range ($< 1000\,\mathrm{Hz}$)
2. Laser frequency might drift. Check timeseries.

- Linewidths are not stable
  1. Check the laser source for instabilities (acoustic, electronic, thermal)
- Noise density values and Linewidths are higher/lower than expected
  1. Check if the digitizer input coupling is properly terminated with $50\,\Omega$. Removing the $50\,\Omega$ termination will increase the output voltage!
  2. Check optical input power. Frequency noise sensitivity scales inversely with optical power
- There are peaks at high frequencies that cannot be caused by laser noise
  The instrument itself has a low noise output without peaks. However, connecting different devices to the PC can cause that noise source from the electronic grid can enter the system.
  1. Please check for ground/earth-loops
  2. Try direct connection between Analyzer and PC
  3. Use different cables (USB, BNC, Ethernet)
- GUI is not working properly / errors occur / crashes
  1. Take screenshots or desktop-videos of the GUI and Analyzer-Front, send/link them to service@highfinesse.de
  2. Include a detailed report of the conditions of the failures
  3. Please also include the version information of the software. These are available from the “Support → About” dialog, which also allows to copy all information into the clipboard.

7 General Information

7.1 Operating conditions

Operation	
Temperature	20 °C – 30 °C
Humidity	35 % – 55 %
Altitude	0 m – 2000 m
Storage	
Temperature	15 °C – 30 °C
Humidity	35 % – 55 %
Altitude	0 m – 2000 m

7.2 Specifications Analyzer Unit

7.2.1 LWA-10k VIS

Input Requirements	
Wavelength range	450 nm – 1064 nm, typical 780 nm
Laser type	CW and single mode
Input power	0.5 mW – 8 mW, typical 5 mW
Input type	PM-FC/APC fiber
Maximum frequency stroke[2)]	40 MHz
Frequency noise	
Noise floor $N_{\Delta\nu}$ at typ. input power and wavelength[3)]	10 Hz: 500 Hz/$\sqrt{\text{Hz}}$,100 Hz: 150 Hz/$\sqrt{\text{Hz}}$,1 kHz: 60 Hz/$\sqrt{\text{Hz}}$,10 kHz: 60 Hz/$\sqrt{\text{Hz}}$,100 kHz: 50 Hz/$\sqrt{\text{Hz}}$,>1 MHz: 30 Hz/$\sqrt{\text{Hz}}$
Frequency noise bandwidth	10 Hz – 10 MHz
Frequency noise density	
Minimum measurable intrinsic linewidth (Lorentzian linewidth)	<12 kHz
Effective linewidth range (optical linewidth)[β-separation method]	<20 kHz-30 MHz
Relative intensity noise	not measurable
Lineshape	
Effective linewidth range (optical linewidth)	<20 kHz-10 MHz
Dynamic Range	60 dB
Miscellaneous	
Power supply	230 V / 115 V ± 10%; 50 Hz / 60 Hz
Power consumption	$<$ 100 W
Interface	Ethernet
Analog output	BNC ± 7.5 V (50 Ω), floating ground

[2)] if “Locked”, during initialization may be lower

[3)] $N_{\Delta\nu}$ is the noise floor of the instrument in terms of the square root of the power spectral density of the frequency noise

Dimensions	$440 \times 340 \times 155\ \mathrm{mm}^3$
Weight	$12\ \mathrm{kg}$

7.2.2 LWA-10k NIR

Input Requirements	
Wavelength range	$1064\ \mathrm{nm}$ – $1625\ \mathrm{nm}$, typical $1550\ \mathrm{nm}$
Laser type	CW and single mode
Input power	$0.5\ \mathrm{mW}$ – $8\ \mathrm{mW}$, typical $5\ \mathrm{mW}$
Input type	PM-FC/APC fiber
Maximum frequency stroke[4)]	$40\ \mathrm{MHz}$
Frequency noise	
Noise floor $N_{\Delta\nu}$ at typ. input power and wavelength[5)]	$10\ \mathrm{Hz}$: $200\ \mathrm{Hz}/\sqrt{\mathrm{Hz}}$,$100\ \mathrm{Hz}$: $100\ \mathrm{Hz}/\sqrt{\mathrm{Hz}}$,$1\ \mathrm{kHz}$: $30\ \mathrm{Hz}/\sqrt{\mathrm{Hz}}$,$10\ \mathrm{kHz}$: $20\ \mathrm{Hz}/\sqrt{\mathrm{Hz}}$,$100\ \mathrm{kHz}$: $15\ \mathrm{Hz}/\sqrt{\mathrm{Hz}}$,>$1\ \mathrm{MHz}$: $10\ \mathrm{Hz}/\sqrt{\mathrm{Hz}}$
Frequency noise bandwidth	$10\ \mathrm{Hz}$ – $10\ \mathrm{MHz}$
Frequency noise density	
Minimum measurable intrinsic linewidth (Lorentzian linewidth)	<$2\ \mathrm{kHz}$
Effective linewidth range (optical linewidth)[β-separation method]	<$5\ \mathrm{kHz}$-$30\ \mathrm{MHz}$
Relative intensity noise	not measurable
Lineshape	
Effective linewidth range (optical linewidth)	<$5\ \mathrm{kHz}$-$10\ \mathrm{MHz}$
Dynamic Range	$60\ \mathrm{dB}$
Miscellaneous	
Power supply	$230\ \mathrm{V}$ / $115\ \mathrm{V}$ ± 10%; $50\ \mathrm{Hz}$ / $60\ \mathrm{Hz}$
Power consumption	$< 100\ \mathrm{W}$
Interface	Ethernet
Analog output	BNC ± $7.5\ \mathrm{V}$ ($50\ \Omega$), floating ground
Dimensions	$440 \times 340 \times 155\ \mathrm{mm}^3$
Weight	$12\ \mathrm{kg}$

7.2.3 LWA-100k NIR

Input Requirements	
Wavelength range	$1064\ \mathrm{nm}$ – $1625\ \mathrm{nm}$, typical $1550\ \mathrm{nm}$
Laser type	CW and single mode
Input power	$0.5\ \mathrm{mW}$ – $8\ \mathrm{mW}$, typical $5\ \mathrm{mW}$
Input type	PM-FC/APC fiber
Maximum frequency stroke[6)]	$40\ \mathrm{MHz}$

[4)] if “Locked”, during initialization may be lower

[5)] $N_{\Delta\nu}$ is the noise floor of the instrument in terms of the square root of the power spectral density of the frequency noise

[6)] if “Locked”, during initialization may be lower

Frequency noise	
Noise floor $N_{\Delta\nu}$ at typ. input power and wavelength[7)]	10 Hz: 200 Hz/$\sqrt{\text{Hz}}$,100 Hz: 100 Hz/$\sqrt{\text{Hz}}$,1 kHz: 30 Hz/$\sqrt{\text{Hz}}$,10 kHz: 20 Hz/$\sqrt{\text{Hz}}$,100 kHz: 15 Hz/$\sqrt{\text{Hz}}$,>1 MHz: 10 Hz/$\sqrt{\text{Hz}}$
Frequency noise bandwidth	10 Hz – 10 MHz
Frequency noise density	
Minimum measurable intrinsic linewidth (Lorentzian linewidth)	<10 kHz
Effective linewidth range (optical linewidth)[β-separation method]	<15 kHz-30 MHz
Relative intensity noise	not measurable
Lineshape	
Effective linewidth range (optical linewidth)	<15 kHz-10 MHz
Dynamic Range	60 dB
Miscellaneous	
Power supply	230 V / 115 V ± 10%; 50 Hz / 60 Hz
Power consumption	$<$ 100 W
Interface	Ethernet
Analog output	BNC ± 7.5 V (50 Ω), floating ground
Dimensions	440 × 340 × 155 mm^3
Weight	12 kg

7.3 Digitizer Unit

Vertical	
Vertical resolution	15 bit
Ranges	±10 mV – ±20 V (11 ranges)
Noise (±10 mV)	70 µV RMS
Flatness	0.3 dB, −3 dB
Horizontal	
Sampling rate (max.)	125 MSa/s
Acquisition time (typ.)	1 ms – 100 ms
Miscellaneous	
Communication	USB 3.0 Type B (USB 2.0 compatible)
Power consumption[8)]	$<$ 5 W
Analog input	BNC ±20 V (1 MΩ), single ended
Dimensions	210 × 200 × 74 mm^3
Weight	2 kg

[7)] $N_{\Delta\nu}$ is the noise floor of the instrument in terms of the square root of the power spectral density of the frequency noise

[8)] Direct USB connection to PC or via external powered USB-Hub

7.4 Software

PC Requirements	
Operating system	Microsoft© Windows© 10 (64 bit) or newer
CPU (recommended)	Intel© i5 8600 / AMD Ryzen™ 5 2600 or better
Memory (recommended)	16 GB
Ports	2 USB3.0 or higher (USB 3.0 recommended) Ethernet / WiFi access to LAN
Software	
Graphical evaluation options	timeseries, frequency noise density, lineshape
Linewidth evaluation	intrinsic lw, effective lw (β-separation and FWHM of the optical spectrum)

7.5 Declaration Information On Disposal at End of Lifetime

HighFinesse Laser and Electronic Systems GmbH (hereinafter referred to as HighFinesse) conforms to the EU legislation for electrical and electronic equipment (EEE), which includes the Waste Electrical and Electronic Equipment Directive (WEEE). Assigned duties affect product design of the equipment, disposal of used appliances as well as organizational responsibilities, i.e. product registration.

There are different requirements for household WEEE and that which is sold business to business (B2B).

All equipment HighFinesse handles is classed as B2B. HighFinesse is registered at the Competent Authority (Stiftung Elektro-Altgeräte Register EAR) under No. DE 14593317.

At end-of-life return your product back to HighFinesse. HighFinesse will dispose used equipment in such a manner as to meet all relevant local, country and EU requirements and guideline.

To return products please mark them clearly with “intended for disposal”. Contact HighFinesse prior to shipping and send them to the following address mentioned in Section 9.

8 Declaration of CE Conformity

CE Konformitätserklärung / Declaration of CE Conformity WS-Series / Declaraction de Conformité

CE

Wir / We / Nous	**HighFinesse Laser and Electronic Systems GmbH**
Anschrift / Address / Adresse	**Neckarsulmer Straße 5** **72072 Tübingen** **Germany**

Erklären in alleiniger Verantwortung, dass das Produkt / declare under our sole responsibility, that the product / declarons sous notre seule responsibilité, que le produit

Bezeichnung / Name / Nom	**LWA-Serie / LWA-Series**
Artikelnr. / Article No. / No. d'Article	**LWA 1k, LWA 10k, LWA 100k**
Beschreibung / Description / Description	**Linewidth Analysator / Linewidth Analyzer**

Mit den grundlegenden Anforderungen der Richtlinien / fulfills the requirements of the standard and regulations of the directives / satisfait aux exigences des normes et directives

2014/30/EU (Elektromagnetische Verträglichkeit), 2011/65/EU (RoHS-Richtlinie)

übereinstimmt und damit den Bestimmungen entspricht. / and therefore corresponds to the regulations of the directive. / et, ainsi, correspond au regiments de la directive.

Angewendete harmonisierte Normen / Applied harmonized standards / Normes harmonisées appliquées:

DIN EN 61326-1	VDE 0843-20-1:2013-07 Elektrische Mess-, Steuer-, Regel- und Laborgeräte — EMV-Anforderungen — Teil I: Allgemeine Anforderungen (IEC 61326-1:2012); Deutsche Fassung EN 61326-1:2013 Electrical equipment for measurement, control, and laboratory use — EMC requirements — Part 1: General requirements (IEC 61326-1:2012); German version EN 61326-1:2013
DIN EN 61010-1	Sicherheitsbestimmungen für elektrische Mess-, Steuer-, Regel- und Laborgeräte — Teil 1: Allgemeine Anforderungen (IEC 61010-1 + COR:2011 + A1:2016, modifiziert + A1:2016/COR1:2019); Deutsche Fassung EN 61010-1:2010 + A1:2019 + A1:2019/AC:2019 Safety requirements for electrical equipment for measurement, control and laboratory use — Part 1: General requirements (IEC 61010-1 + COR:2011 + A1:2016, modified + A1:2016/COR1:2019); German version EN 61010-1:2010 + A1:2019 + A1:2019/AC:2019
DIN EN 63000	Technische Dokumentation zur Beurteilung von Elektro- und Elektronikgeräten hinsichtlich der Beschränkungen gefährlicher Stoffe; Deutsche Fassung EN 63000:2018 Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances; German version EN 63000:2018

Durch nicht von uns autorisierte Änderungen am Produkt verliert diese Erklärung ihre Gültigkeit. / Non authorized changes at the product result in the invalidity of this declaration. / Changements au produit, qui ne sont pas autorisées par nous, ont pour conséquence l'invalidité de cette declaration.

Tübingen, den 01.01.2025

Dr. Thomas Fischer
CEO & President

Thomas Fischer

Ort und Datum der Austellung
Place and date of issue
Lieu et date d'établissement

Name und Unterschrift des Befugten
Name and signature of authorized person
Nom et signature de la personne autorisée

9 HighFinesse Information / Service

HighFinesse
The Standard of Accuracy

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Document version 2025-06 0007

10 Changelog LWA-10k/100k Manual

Document Version 2024-06 0002:

- Updated CE declaration (ch. 8, p. 38)
- Added troubleshooting hints for “failed” initialization. (ch. 6, p. 32)

Document Version 2025-02 0006:

- LWA-10k/100k series instruments do require polarization maintaining (PM) fiber, single mode (SM) is not sufficient. (ch. 7, p. 34)